

A frozen read-out for early Barrett neoplasia: one positive result, a set of null results, and a domain gap that reproduces

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Code: <https://github.com/liyifreddy/rare26-frozen-readout>

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Abstract

We report a submission to the RARE26 challenge, which asks for detection of early neoplasia in Barrett’s esophagus and ranks entries by positive predictive value at 90% recall with prevalence fixed at 1%. The training set contains 158 positive images, which rules out fine-tuning a large encoder, so we froze the backbone entirely and worked on the layer above it. The only learned parameters are one discriminant vector and one mean vector, 4096 numbers in total. One change earned its place: scoring each of the 49 spatial positions separately and taking the top 2%, rather than averaging the map first. It is detectably positive in both cross-center directions on all three read-outs and survives five further checks, including an external localization test against five independent expert delineations. Almost nothing else was. Ten alternative backbones, seven read-outs, the choice of layer, and every component combination we tried produced no configuration that is detectably better in both directions, and the improvements that do exist all land on backbones other than the one we ship. On the challenge validation set the pipeline reaches AUROC 0.7700 and PPV@90recall 0.0152, against a random baseline of 0.0100 and our own cross-center AUROC of 0.9605 and 0.9883. A team in the previous edition of this challenge, using a different architecture, reported the same pattern and almost the same number: cross-validation ROC-AUC 0.9340 to 0.9600 falling to 0.7709 externally. We simulated three acquisition differences and found they account for a small fraction of that gap. The rest is not in the imaging. Most of what we learned is in the evaluation protocol rather than in the method, so we describe it in more detail than the results it supports, and we list the mistakes it failed to catch.

How to read this report

Every number here is inserted by a script, not typed by hand. The repository holds the table that maps each one to its source, together with the code that generates it. Most of those sources are result files in the repository; the rest are the published classifier, the fitting code, or something outside it, and the repository lists that last group by name so you can see which numbers you cannot check there. Where a quantity was measured but its uncertainty was not, the table says so and leaves the interval blank rather than borrowing one from a different estimator.

1. The task, and what the metric actually measures

The task is binary. Given an endoscopic image of Barrett’s esophagus, decide whether it contains early neoplasia. The platform delivers one case as a single stacked file of 384 frames and expects one score per frame.

Ranking uses a single number: positive predictive value at 90% recall, with prevalence fixed at 1%. Everything in this report follows from the shape of that number, so it is worth writing it out. At fixed prevalence,

$$PPV = 0.9 \pi_i / [0.9 \pi_i + (1 - spec) (1 - \pi_i)], \quad \pi_i = 0.01$$

where spec is specificity at the operating point where sensitivity is 90%. PPV is a strictly increasing function of that specificity. The two cannot point in opposite directions. They can only differ in how noisy they are.

That is the easy half. The harder half is where the operating point comes from. To reach 90% recall you place the threshold at the score of the $\text{ceil}(0.9 * n_{\text{pos}})$ -th lowest positive. That is one order statistic of the positive scores, not an average over anything. With 158 positives in the training set and fewer in any single evaluation fold, that order statistic moves whenever a handful of hard positives move.

The prevalence then does the rest. At 1% prevalence, once the threshold is set, the value of PPV is decided by how many negatives sit above it, which is the far tail of the negative score distribution. A small shift in the threshold sweeps a lot of that tail.

So the ranking metric is a threshold statistic that reads off a single point, while AUROC is an average over all positive-negative pairs. They are not two views of the same thing at different resolutions. The following table gives the correspondence under a binormal model with equal variances, and includes the random baseline.

AUROC	implied spec@90	implied PPV@90R at 1% prevalence
0.50	0.100	0.0100 (random baseline)
0.65	0.231	0.0117
0.77	0.406	0.0151
0.84	0.550	0.0198
0.93	0.790	0.0414
0.96	0.884	0.0726
0.99	0.978	0.2896

Two consequences run through the rest of this report. First, an AUROC that looks respectable can sit on a PPV that is close to the random baseline. Second, anything that disturbs a small number of hard positives will show up in the ranking metric long before it shows up in AUROC. We report both numbers side by side everywhere, and when they disagree we treat the disagreement as the result rather than choosing one.

2. The data, and what it is not

The public training set has 3095 images, of which 158 are positive, collected from two centers. Center 1 contributes 2279 images with 61 positives; center 2 contributes 816 images with 97 positives.

Both centers are Dutch, both collections are retrospective, and the challenge documentation states that they were acquired without a standardized imaging protocol. The validation and test cohorts are different in three ways at once. They draw on twelve centers rather than two. They include prospectively acquired images. And those prospective images follow a standardized protocol, acquired by endoscopists specializing in Barrett surveillance.

This matters more than a sentence in a limitations section. The strongest evaluation we can build from the public data is to train on one center and test on the other, in both directions. That measures robustness to the difference between two retrospective Dutch collections. It does not measure robustness to the difference between a two-center retrospective collection and a twelve-center partly prospective one. **The shift we can validate is not the shift we are evaluated on.** Every internal number in this report should be read with that sentence attached.

Labels are the second gap. The training annotations were produced without the involvement of a specific expert group, while the prospective part of the evaluation data was acquired by Barrett specialists. We did not have access to the evaluation labels, so we cannot measure the difference directly. We can measure the scale of expert disagreement on a comparable task: on an independent Barrett dataset where five experts delineated the same lesions, the median pairwise intersection over union between two experts was 0.7047 across 500 pairs.

Finally, a note on quantity. 158 positives is the number that determines what can and cannot be established here, and it appears again in section 5 when we describe how wide the confidence intervals turn out to be.

3. Why the encoder is frozen

With 158 positive images, fine-tuning a large encoder is not the right use of the data. The binding constraint is the quality of the representation, not the capacity to fit. A network with tens of millions of free

parameters and a few hundred positives will find the training set long before it finds the disease.

We therefore froze the encoder completely and put the work above it. Nothing in the backbone is trained. The learnable parameters are one discriminant vector and one mean vector, 4096 numbers in total, about 16 KB on disk. This makes every result in section 6 an experiment about the read-out and the scoring layer rather than about optimization.

Two boundary conditions are worth stating, though neither is the reason for the choice. The pretrained weights come with a data use agreement that permits academic research and forbids redistribution, so they are declared but not shipped with our code. Compute was a single laptop GPU, which rules out repeated pretraining runs but not the experiments we actually ran, since the encoder is frozen and features are extracted once.

The design has a side effect that turned out to be useful. Because the encoder never changes, every ablation is a change to a small closed-form computation on cached features. That made it cheap to run each comparison in both cross-center directions, with paired resampling, rather than once with a single split.

4. Related work

Three lines of prior work bear on this report. We use them to place our numbers rather than to justify the method.

The previous edition of this challenge. RARE25 ran the same task with the same metric and published a summary. Two facts from it matter here. The organizers note that “although several methods achieved strong overall discriminative performance, positive predictive values remained low for most approaches, emphasizing the intrinsic difficulty of low-prevalence detection”. And one team reported cross-validation ROC-AUC between 0.9340 and 0.9600 that fell to ROC-AUC 0.7709 and PPV@90recall 0.0112 on a separate development set. The winning entry reached PPV@90recall 0.3200 on the full closed test set, but a median of 0.0350 under resampling at the intended class imbalance. We return to these numbers in section 8, because our own results land on top of them.

Label noise and the choice of metric. Menon et al. prove that under class-conditional label noise the corrupted AUC is a positive affine transform of the clean AUC, so the ranking induced by any scorer is unchanged. They also state the limit of that result directly: for measures other than balanced error that are optimized by thresholding the class probability, one has to know the noise rates or the base rate in order to place the threshold. Precision at a fixed recall is such a measure, and they do not analyze it. Elkan and Noto give the corresponding correction for precision when positives are contaminated in the positive-unlabelled setting. What is missing is not the theory but the magnitude: how much does a fixed amount of label noise cost a fixed-recall precision metric relative to what it costs AUC. Section 6 reports that measurement.

Acquisition differences in endoscopic AI. The provider of the pretrained weights has published on the effect of endoscope enhancement settings on the performance of endoscopic AI systems, and on the evaluation of a Barrett CAdE system under real-world imaging conditions. We take the mechanism from that work and test it directly in section 8, where we perturb color, sharpening and compression and measure how much of our own gap each can account for.

5. How the evaluation protocol was set

This section is longer than the results section it supports. That is deliberate. Most of what we learned in this project is in the protocol rather than in the method.

Both directions, always. Every comparison is run twice: train on center 1 and evaluate on center 2, then swap. A result in one direction is never adopted. This is not a robustness flourish. In our data the two directions disagree often enough that single-direction results are close to worthless, and we report below how often.

Paired resampling. Differences are estimated by a paired bootstrap that resamples cases within patient groups and resamples both classes. Comparing two configurations on the same resample removes the variance that dominates when each is bootstrapped separately.

Grouping. Consecutive frames of one examination are near duplicates. If they straddle a split, the evaluation measures memorization. We group near-duplicate images before any split, using mutual nearest neighbors with a cross-center constraint, so that one examination cannot appear on both sides.

Two keys, reported together. AUROC and the ranking metric are reported side by side for every comparison. AUROC is the lower variance estimate and we use it to screen. The ranking metric is what decides adoption because it is what the leaderboard uses. When the two disagree, we report both and treat the disagreement as the finding. Section 6 contains several cases where reporting only one key would have converted a measured result into a non-result, and one where it did.

A signed verdict with seven classes. Each comparison is classified from the two directions together, not by eyeballing point estimates:

Class	Meaning	Condition
A+	established (improvement)	Both directions detectable and positive, and both at or above the threshold
A−	established (degradation)	Both directions detectable and negative, and both at or above the threshold
B+	directional, below practical threshold	Both directions detectable and positive, at least one below the threshold
B−	directional, below practical threshold	Both directions detectable and negative, at least one below the threshold
C	ruled out	Both intervals lie wholly inside the threshold band
D	tested, not detected	Anything else
E	direction conflict	Directions disagree in sign, at least one detectable and large

Threshold: $|\Delta \text{AUROC}|$ of 0.0165, measured from a power curve on this data. Detectable means the paired bootstrap interval excludes zero.

The important detail is the sign constraint. An early version of this classifier required only that both directions be detectable and large enough, without requiring them to agree in sign. It labeled configurations that were reliably *worse* as established. That mistake is easy to make and hard to see in a summary table, which is why the classifier is a single function that every script calls rather than a rule each script applies.

A detection threshold that was measured, not chosen. The threshold for calling a two-direction difference detectable is 0.0165 on the AUROC scale. It comes from a power curve measured on this data, not from a convention.

Pre-registration. Before any grid was run, the decision rule for reading its output was written down and dated: which quantities would be reported, what would count as adoption, and what would be done if the result was ambiguous. The rules are in the repository with their timestamps. Writing them afterwards would have let the numbers choose the rule.

No argmax, and the control that justifies it. We never take the best cell of a grid. The reason is measured rather than assumed. We ran one strictly nested search in which an inner loop enumerated 120 configurations per backbone per direction and kept the best, and an outer fold that the inner loop never saw reported the result. On the ranking metric, 9 of 22 paired cells were detectably worse and 1 was better. The mechanism is ordinary: the maximum of N noisy estimates is inflated by about σ times the square root of $2 \ln N$, which is 3.1 σ at $N = 120$, far above anything this dataset can resolve.

What the protocol cannot do. All of the above controls selection and noise. None of it controls the gap described in section 2. A protocol can only be honest about the distribution it samples from, and ours samples from two retrospective Dutch centers.

6. What we tried, and what each thing did

Experiment family	Backbones	Cells	A+	A−	B±	C	D	E	double-key conflicts	Verdict	Source
Read-out basis x component (E3)	11	474 (96 dup removed)	39	14	28	101	259	33	86	39 improve in both directions, 14 degrade	e3_basis_component.json
Layer 3+4 replication (E3b)	8	192 (96 dup removed)	0	7	5	32	148	0	43	none improve, 7 degrade	e3b_cnn_layer.json
Backbone x module interaction	4	48	0	0	9	23	16	0	-	none improve	b1_backbone_module.json
Field-of-view mask	4	16	0	0	0	14	2	0	-	none improve	c2_fov_mask.json
Fit-scope arm D	4	16	0	4	2	0	9	1	-	none improve, 4 degrade	c4_arm_d.json
Instance selection (MIL)	11	33	3	0	0	7	22	1	-	3 improve in both directions	r5_mil.json
Layer choice	11	11	1	0	1	0	9	0	-	1 improve in both directions	r4_layer.json
Same-backbone head fusion	11	99	0	25	31	32	9	2	-	none improve, 25 degrade	r6_fusion.json
Multi-backbone fusion	-	10	0	0	1	3	6	0	-	none improve	r7_bbfusion.json
TTA and color normalization	10	30	0	1	0	15	12	2	-	none improve, 1 degrade	r8_tta_color.json
External covariance target	-	27	0	4	9	13	1	0	-	none improve, 4 degrade	r2_extcov.json
One-class vs discriminant	11	44	0	40	0	0	4	0	-	none improve, 40 degrade	q1_oneclass_all.json
Geometric read-outs	8	32	0	4	4	8	15	1	1	none improve, 4 degrade	q3_geometry_all.json
Greedy combination search	11	11	1	3	0	0	2	5	-	1 improve in both directions, 3 degrade	r11_combo.json
Normalization family	11	176	0	33	9	89	43	2	-	none improve, 33 degrade	d2_norm_family.json
All backbones, A-class components	11	616	17	51	14	397	124	13	-	17 improve in both directions, 51 degrade	d1_all_backbones.json

Not included, and why:

- `c1_pooling_ops.json` (Pooling operator x backbone): no delta field

The table above is the whole record. This section says what the rows mean.

One change survived. Replacing global average pooling with per-position scoring followed by top-2% pooling is the only modification that is detectably positive in both directions on all three read-outs. Under true cross-center evaluation it moves AUROC by +0.0486 and +0.0163, specificity at 90% recall by +0.2069 and +0.0682, and the ranking metric by +0.0512 and +0.4169. Residual false positives at 90% sensitivity fall by 66.9% and 89.3%.

We then tried to break it five more ways. The gain concentrates in the hard positives and is near zero on the easy stratum, which is at ceiling, so it improves the images that set the threshold rather than pushing easy cases higher. On an independent Barrett dataset where five experts delineated lesions separately, the highest-scoring cell falls inside the expert consensus lesion 78.0% of the time against 16.7% for a random cell, and the lift grows with the level of expert agreement. Repeating the entire component grid on a different layer gives 192 cells with 0 improvements. A two-direction null control puts the joint chance rate at 0.7%, measured rather than obtained by multiplying the per-direction rates. And the offline reproduction matches the container’s own output to within 4.917e-07 per image with 0 rank changes.

Component improvements exist, and none of them is on the backbone we ship. Across every family in Table 3, 61 cells improve in both directions. All but 1 of them are on a ViT, and the exception is on out-of-domain CNN weights. On the delivered backbone the count is 0. Ten alternative backbones, seven read-outs and the choice of layer produced no configuration that is detectably better than the delivered one in both directions. That is a statement about this dataset as much as about the alternatives: at 61 and 97 positives the paired intervals on the ranking metric are wider than the differences being tested.

The two keys disagree often enough to matter. In the largest grid, the number of cells where AUROC and the ranking metric point in incompatible directions is 86 out of 474. A protocol that reports one key would present about one comparison in five as settled when it is not. This is the same effect described in section 1, seen from the other side: the two numbers summarize different things, and at this sample size the difference is not academic.

One family stands out for being decisive rather than inconclusive. Modeling the normal class alone and scoring by distance from it, rather than fitting a discriminant between the two classes, is detectably worse in both directions in 40 of 44 cells. With 158 positives it is tempting to treat the task as anomaly detection; on this data that choice costs a great deal, and unlike most of our comparisons the cost is measurable.

The assembly procedure is determined by where it starts. Building the configuration one component at a time, with each component required to clear the pre-registered two-direction threshold, leaves every one of 11 backbones exactly where it began, from either starting point. The number of backbones on which the two starting points converge is 0. We report this as a negative result about the procedure, not as support for the configuration we happened to start from.

Label noise costs the ranking metric one to two orders of magnitude more than it costs AUC. We flipped labels on the training set at increasing rates, in two ways: uniformly within each class, which is the setting Menon et al. analyze, and preferentially on the hardest positives, which is closer to how annotators actually disagree. Training used the flipped labels; evaluation always used the true ones. At a 5% flip rate the ranking metric falls by 30.5% to 46.0% while AUROC falls by 0.3% to 2.5%, a ratio of about 19 times to about 106 times across four curves. This is the empirical complement to the theory cited in section 4, and it is the clearest single demonstration of the point made in section 1.

7. What we submitted

The delivered pipeline is the frozen backbone, the per-position discriminant, and top-2% pooling. In full:

Pipeline step	What it does	Where the evidence is
Read the stacked file Resize 512 then 224	One .tiff or .mha per case, N frames of 512x512 Two downscales, both anti-aliased, matching the training path	p28_384.json, p29_platform.json t7_train_serve.json, audit_2stage.json
Frozen RN50 layer4	GastroNet-5M DINOv1 weights, no fine-tuning, 7x7x2048 kept	d1_all_backbones.json, r4_layer.json
Signed power transform	sign(x) times x ^0.5, applied before the head	e3_basis_component.json

Pipeline step	What it does	Where the evidence is
Shrinkage discriminant	49 positions as samples for the within-class scatter, lambda by grouped CV on pAUC(85-95)	lambda_paired.json, lambda_endpoint.json
Top 2% pooling	Score every position, take the maximum on a 7x7 grid	p23_cross_confirm.json, c1_pooling_ops.json
Localization check	Highest-scoring cell against five-expert consensus on an external set	r9_evc_loc.json
Logistic squash	Monotone, so it cannot change the metric	scorer.py --demo self-test 1
Container output	One score per frame, in frame order	p29_platform.json

The shrinkage coefficient is selected by repeated grouped cross-validation on partial AUROC, and the final head is fitted on all images from both centers. Deployment is a CPU-only container. One case is one file of 384 frames with a 600.0 second limit; measured end to end from container start, including model loading, a case takes 15.8 seconds on a laptop CPU and 26.0 seconds on the evaluation hardware (ml.r7i.large (2 vCPU, 16 GB, no GPU)). Peak memory lies between 2 GB and 3 GB against 16 GB configured.

The container was verified by reloading the packaged image, running it with the network disabled, and comparing its output to the offline computation. On the platform itself, the same stack scored through the platform's own ingestion produced 0 rank changes against the local run, with a Spearman correlation of 1.0000.

The weights themselves are not ours to hand out. They are covered by the *GastroNet-5M Dataset and Model Weights – Unified Data Use Agreement (Academic & Commercial R&D) v1.0*, which is free for academic use and forbids redistribution. The submission container carries a copy of the file, because the evaluation runs with the network disabled and the challenge submission format requires a self-contained image. Our public repository does not. Anyone reproducing this work obtains the same file from the provider's pretrained-model listing at <https://cortex.thetavision.nl/dataset-provider/listing/2/>, accepts the agreement, and places it at the path the repository documents. That listing covers eight weight files under one agreement; ours is the second, ResNet-50 pretrained with DINOv1 on GastroNet-5M. No layer of it is fine-tuned.

8. What happened on the held-out data

Evaluation	PPV@90R at 1%	spec@90	AUROC
Our cross-center, train c2	0.0787	0.894	0.9605
Our cross-center, train c1	0.5214	0.992	0.9883
RARE25 validation	0.0151 [0.0106, 0.0345]	0.407	0.8430 [0.7121, 0.9621]
RARE26 validation	0.0152 [0.0104, 0.0306]	0.411	0.7700 [0.6354, 0.8968]
Random ranking	0.0100	0.100	0.5000

Intervals on the two validation rows are the challenge evaluator's own bootstrap. The cross-center rows carry none: what was recorded there is the interval on the paired difference against global average pooling, which is a different quantity. The two validation rows are the 2026-08-31 submission on the Open Development Phase leaderboard, <https://rare26.grand-challenge.org/evaluation/open-development-phase/leaderboard/>, recorded as read on that date because the page carries several submissions per entrant and reorders as others submit.

On the challenge validation set the pipeline scores AUROC 0.7700 with a 95% interval of [0.6354, 0.8968], and PPV@90recall 0.0152 against a random baseline of 0.0100. On the RARE25 validation set the same container scores AUROC 0.8430 and PPV@90recall 0.0151. Our own true cross-center AUROC is 0.9605 and 0.9883.

The first thing to check is whether the pipeline is working at all. It is. Inverting the identity of section 1 gives an implied specificity at 90% recall of 0.4110, and a binormal model with AUROC 0.7700 predicts almost exactly that value. The two numbers are consistent with each other, and both are well above the random baseline. Had the frame ordering or the label polarity been wrong, AUROC would have sat at 0.5.

The second thing is that this is not specific to us. The team quoted in section 4 reported cross-validation ROC-AUC of 0.9340 to 0.9600 and an external ROC-AUC of 0.7709 with PPV@90recall 0.0112. Our figures are 0.7700 and 0.0152. The drop reproduces across teams, methods and editions, which makes it a property of the task rather than a fault in any one pipeline.

We then tried to attribute the gap. Three acquisition axes were simulated on the test side while the head stayed fitted on unperturbed training data: color and illumination, unsharp masking, and compression artifacts. Each was pushed well past any realistic setting.

The result depends on which key you read it with, and the disagreement is itself informative. On AUROC, at realistic magnitudes the three axes together account for about 15.7% of the gap in one direction and 0.9% in the other, and even at implausible strength color alone accounts for at most 28.8% and 8.1%. On the ranking metric the same calculation returns 129.5% and 146.1%.

We do not read the second pair as showing that imaging explains half the gap. The ranking metric responds this strongly to any perturbation, which is the property established in section 1 and measured again in section 6, and on the main metric the three shares sum to more than the whole, which is direct evidence that they are not a decomposition. A quantity that moves a great deal in response to everything cannot attribute anything to one cause. The attribution is therefore identifiable on AUROC and not identifiable on the ranking metric, and what it says on AUROC is that these three acquisition differences are not the main driver. That conclusion carries the caveat that AUROC is the less sensitive of the two keys, which is why we state it as a bound rather than as a decomposition.

A measurement taken alongside it points the same way. Unsharp masking displaces the features by 49.5% of their norm while changing AUROC by -0.0013 and -0.0036. The features move a long way and the score does not follow. Measured directly, the fraction of the discriminant's energy lying in the illumination subspace is 0.2% and 0.2%.

On the evidence that is identifiable, what remains is not imaging. Section 2 named the two candidates we cannot test: the annotation was produced by different people under a different protocol, and the case mix spans twelve centers with a prospective component rather than two retrospective collections. Neither is something a pre-processing step can repair, and we hold no images from the target distribution with which to learn one.

We also tested four modules intended to buy robustness to acquisition differences. Two are detectably worse in one direction. The other two are not detectably worse on either key, but their point estimates on the ranking metric are negative in both directions, and the intervals are wide enough that “not detectably worse” carries very little information here: across eight cells, the number in which the ranking metric detects any difference at all is 0. We did not adopt any of them. Their code and their numbers are in the repository.

9. What we would do differently

Get one measurement from the target distribution before choosing the question. We spent most of the available time on backbones, read-outs and component combinations, all validated on a split between two retrospective Dutch centers. The first number from the platform showed that the split we were validating on was not the split we were being scored on. Had that number arrived in the first week, the work would have gone into the question of what changes across centers rather than into which frozen encoder to use.

Treat the metric as a threshold statistic from the start. Section 1 states the identity we used throughout, but we derived its consequences late. The single order statistic at the heart of the metric explains the label-noise result, the size of the confidence intervals, and why almost nothing we tried could be distinguished. A protocol built around that fact from day one would have set different thresholds and asked for different sample sizes.

Build gates that can fail. Six checks in this project reported success without having examined anything: a verification script that exited zero when the container engine was not running, a repository scanner that reported all clear after matching no files, a placeholder check that recognized one of the two markers actually present in the text, a registration block that wrote nothing and exited zero, an adoption rule that passed on a direction with nothing in it to separate, and a check that read its own success as failure because the command it used prints a matching line for negation rules too. The first four were found by accident. The last two were found on purpose, once we started running every gate twice: once against a deliberately broken input to prove it reports red, then once for real. A gate that cannot report failure is worse than no gate, because it is mistaken for evidence. The errors from this project that still matter to a reader are in the repository at docs/05_what_we_got_wrong.md.

Write the reading rule before the numbers. This one worked, and it is the practice we would keep un-

changed. Every grid in this project has a dated note stating what would count as adoption before the grid was run. On more than one occasion the result was tempting and the note was the only thing standing between us and a decision the data did not support.

Declarations

Intended use. The pretrained weights are licensed for research and development only. Section 11 of the agreement states that they are not a certified medical device and have not been clinically validated. Nothing in this report is a claim of clinical readiness. The system was built for a challenge, fitted on challenge training data, and evaluated on challenge validation data; it has not been tested in any clinical workflow, on any prospective cohort of our own, or by any reader study.

Data. The training and validation images belong to the challenge organizers and are not redistributed here. No image from the challenge data appears in this report or in the public repository.

Code and numbers. The code, the fitted head, the classifier that produces every verdict in these tables, the pre-registered reading rules, and the result files are at <https://github.com/liyifreddy/rare26-frozen-readout> under the MIT license. No number here was typed by hand; each is inserted by script from a table that names its source. Most of those sources are result files in that repository. The rest are the published classifier, the fitting code, or something outside it — the challenge’s own job page, the RARE25 summary paper, an internal working note — and the repository lists those keys by name so a reader can see which numbers they cannot check there. The pretrained weights are excluded, for the reason given in section 7.

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References 1 to 5 are required by section 4 of the data use agreement covering the pretrained weights, which asks that the dataset and every publication listed on the provider’s page be cited.